When Depth Signals Collapse: Architecture-Conditional Response in Recursive LLM Generation Chains

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Abstract

Recursive LLM generation—where model outputs are fed as inputs to subsequent generation steps—creates text of varying provenance depth. We investigate whether distributional features can estimate generation depth beyond binary real-vs-synthetic classification. Through experiments across 5 generators from 3 model families (Pythia, OLMo, Qwen), we find that depth signal is *not* a universal property of recursive generation but is strongly **architecture-conditional**: Pythia-family generators exhibit robust depth-correlated feature drift (Spearman $\rho = -0.61$ for entropy_mean), while OLMo and Qwen generators show weak or absent signal ($\rho = -0.22$ and -0.18 respectively). Surface diversity features (TTR, hapax ratio) dominate reference-model entropy features across all architectures, and fine-grained ordinal classification (5-class) is infeasible beyond the gen-0 boundary (macro F1=0.38). Paraphrastic and continuation chains exhibit sign-reversed depth responses with near-zero cross-transfer (F1=0.38–0.42), demonstrating that "depth signal" is process-specific rather than unified. We argue that the *pattern of signal presence and absence* itself constitutes a second-order generator fingerprint, and establish empirical boundaries on when depth estimation is feasible. We position these findings as a characterization study with implications for data provenance research.

1 Introduction

Large language models are increasingly used to generate training data, summaries, and web content, creating feedback loops where model outputs become inputs to subsequent generation Shumailov et al. (2024); Alemohammad et al. (2024). Understanding the *depth* of such recursive generation—how many times text has been processed through LLMs—is important for data governance and contamination assessment.

Prior work on LLM-generated text detection has focused exclusively on binary classification: real vs. synthetic Mitchell et al. (2023); Hans et al. (2024). The question of *ordinal depth estimation*—distinguishing gen-1 from gen-2 from gen-3+—remains unexplored. A natural hypothesis is that recursive generation induces monotonic distributional drift detectable via reference-model statistics, enabling depth estimation as a general-purpose tool.

In this work, we test this hypothesis and find it **does not hold universally**. Through systematic experiments across 5 generators from 3 architectural families, we demonstrate that:

1. Depth signal strength is **architecture-conditional**: strongly present for Pythia-family models (binary LOGO F1=0.78–0.80) but near-chance for OLMo and Qwen (F1=0.57–0.62).

2. Surface diversity features (TTR, hapax ratio) carry most of the discriminative signal; reference-model entropy features add negligible value in cross-family settings (McNemar $p = 0.12$, NS).

3. Fine-grained depth estimation (5-class) is infeasible: macro F1=0.38, with depths 2–4 nearly indistinguishable (Cohen's $d < 0.12$).

4. Paraphrastic and continuation chains exhibit **sign-reversed** feature-depth correlations with near-zero cross-transfer, precluding a unified depth model.

Rather than viewing these as purely negative results, we reframe them as *characterization findings*: the pattern of where depth signal emerges versus collapses constitutes an architecture-conditional fingerprint. A generator's depth response profile—whether features drift, remain stationary, or reverse direction—encodes information about the generator's architecture and training even when absolute depth estimation fails.

Our contributions are: (1) the first systematic study of ordinal depth estimation feasibility across diverse LLM architectures; (2) empirical identification of conditions under which depth signal exists vs. collapses; (3) evidence that signal absence patterns function as second-order generator fingerprints; (4) practical boundaries on depth estimation granularity.

2 Method

2.1 Generation Chain Construction

We construct recursive generation chains of depth 0–4 using 5 base-model generators: Pythia-2.8B, Pythia-1B, OLMo-2-1B, Qwen3-1.7B, and Qwen3-0.6B. Each chain starts from a human-written seed (gen-0) and iteratively produces continuations. We construct both *continuation chains* (open-ended generation) and *paraphrastic chains* (meaning-preserving rewrites) to compare depth dynamics under different generation regimes.

2.2 Feature Extraction

We extract 7 features per text using Pythia-410M as a reference model:

Surface diversity features (2): Type-token ratio (TTR) and hapax legomena ratio (hapax_ratio), measuring lexical diversity without reference-model dependence.

Reference-model entropy features (5): Mean, standard deviation, minimum, and maximum of per-token log-probability under the reference model, plus tail decay rate (tail_decay) measuring the heaviness of the low-probability tail.

2.3 Depth Classification

We evaluate classification at two granularities:

- **3-class**: gen-0 / gen-1 / gen-2+ (coarse provenance)
- **5-class**: gen-0 / gen-1 / gen-2 / gen-3 / gen-4 (fine ordinal)

Primary classifier: XGBoost with 5-fold cross-validation. Generalization is measured via Leave-One-Generator-Out (LOGO) evaluation. We additionally evaluate ordinal-specific methods (CORAL, Frank-Hall) and an end-to-end QLoRA fine-tuned Qwen3-0.6B classifier.

2.4 Feature Dynamics Classification

For each feature, we compute Spearman ρ between feature value and generation depth across all samples. Features are classified as:

- **Drifting**: $R^2 \geq 0.85$ on aggregated depth means (strong monotonic trend)
- **Stationary**: $R^2 < 0.35$ (no systematic depth dependence)
- **Ambiguous**: $0.35 \leq R^2 < 0.85$

3 Experiments

3.1 Experiment 1: Feature Ablation and Trend Analysis (Pythia-only)

We first establish baseline depth signal on same-family data (Pythia-2.8B generator, Pythia-410M reference).

Feature Set	Macro F1	95% CI
Full 7-feature	0.81	[0.79, 0.83]
Surface only (TTR + hapax)	0.77	[0.75, 0.79]
Reference only (5 entropy)	0.68	[0.65, 0.71]
entropy_mean alone	0.64	[0.61, 0.67]

Table 1: Feature ablation on Pythia-2.8B (5-fold CV, 3-class). Surface features alone achieve 0.77 F1, outperforming reference-only by 9pp (bootstrap $p = 0.003$).

Feature	Slope	R^2	Class
hapax_ratio	-0.14	0.93	Drifting
ttr	-0.11	0.91	Drifting
entropy_mean	-0.42	0.88	Drifting
entropy_min	-0.09	0.42	Ambiguous
entropy_max	-0.06	0.31	Stationary
tail_decay	-0.02	0.15	Stationary
entropy_std	$+0.01$	0.08	Stationary

Table 2: Linear trend per feature across depths 0–4 (Pythia-only). Drifting features show strong monotonic decline; stationary features show negligible depth dependence.

On same-family data, depth signal is clearly present. Three features exhibit strong monotonic drift with depth (hapax_ratio, TTR, entropy_mean; $R^2 \geq 0.88$), while three remain stationary (entropy_max, tail_decay, entropy_std; $R^2 \leq 0.31$). Notably, surface features alone (0.77) outperform reference features (0.68) by 9pp ($p = 0.003$), indicating that vocabulary impoverishment—not distributional structure shift—is the primary signal carrier even in this best-case scenario. Perplexity shows strong negative correlation with depth (Spearman $\rho = -0.95$), confirming that deeper generations are more predictable to the reference model.

3.2 Experiment 2: Cross-Generator Evaluation

We evaluate generalization across 5 generators via LOGO cross-validation on 18,200 total samples.

Held-out Generator	3-class F1	Binary F1	Family
Pythia-2.8B	0.82	0.78	Pythia
Pythia-1B	0.84	0.80	Pythia
OLMo-2-1B	0.63	0.58	OLMo
Qwen3-1.7B	0.67	0.62	Qwen
Qwen3-0.6B	0.64	0.57	Qwen
Average	**0.72**	**0.67**	—

Table 3: LOGO cross-generator results. A clear two-tier pattern emerges: Pythia generators (F1$\geq$0.78) vs. non-Pythia (F1$\leq$0.62). Per-generator F1 std=0.11.

The universality hypothesis collapses: a stark two-tier pattern emerges. Pythia-family generators show strong depth signal (held-out binary F1=0.78–0.80), while OLMo and Qwen generators fall near

Feature Set	Avg LOGO F1	p vs Full
Full 7-feature	0.72	—
Surface only	0.70	0.12 (NS)
Reference only	0.61	< 0.001

Table 4: Feature ablation under LOGO. Full features do not significantly outperform surface-only (McNemar $p = 0.12$), but both significantly outperform reference-only.

chance (F1=0.57–0.62). Leave-one-family-out confirms this: held-out Pythia achieves 0.79, held-out OLMo 0.58, held-out Qwen 0.60.

Feature-level analysis reveals the mechanism: entropy_mean Spearman correlation with depth is -0.61 for Pythia-2.8B but only -0.22 for OLMo-2-1B and -0.18 for Qwen3-0.6B—a 2.8$\times$ effect size reduction. The full 7-feature set provides no significant advantage over surface features alone (0.72 vs 0.70, $p = 0.12$), indicating that reference-model entropy features—computed under Pythia-410M—fail to capture cross-family depth dynamics.

Generator	ρ	p-value	Effect
Pythia-2.8B	-0.61	< 0.001	Strong
Pythia-1B	-0.55	< 0.001	Moderate-Strong
Qwen3-1.7B	-0.31	< 0.001	Weak-Moderate
OLMo-2-1B	-0.22	< 0.001	Weak
Qwen3-0.6B	-0.18	< 0.001	Weak

Table 5: Spearman ρ between entropy_mean and depth by generator. Despite statistical significance (large N), non-Pythia effect sizes are too small for practical discrimination.

3.3 Experiment 3: Paraphrastic vs. Continuation Chains

We compare depth signal properties across two generation regimes: paraphrastic chains ($N = 4800$) and continuation chains ($N = 4600$).

Metric	Para.	Cont.	Δ
3-class LOGO F1	0.60	0.72	-0.12
Binary LOGO F1	0.53	0.67	-0.14
ρ(entropy_mean, depth)	$+0.15$	-0.42	reversed
ρ(TTR, depth)	$+0.31$	-0.48	reversed
ρ(hapax, depth)	$+0.28$	-0.51	reversed
Depth retention	0.92	0.83	$+0.09$

Table 6: Paraphrastic vs. continuation chains. Sign reversal across all features demonstrates fundamentally different depth dynamics.

The two chain types exhibit **sign-reversed** feature-depth correlations: continuation chains show monotonic feature decline with depth (as expected from distributional drift), while paraphrastic chains show weak positive correlations consistent with attractor convergence theory Bertrand et al. (2025). Paraphrastic features converge by generation 3.

Cross-transfer F1 is near chance (0.38–0.42), demonstrating that models trained on one chain type cannot classify the other. This rules out any unified "depth signal" and establishes that depth dynamics are fundamentally process-dependent. Paraphrastic binary F1 of 0.53 is effectively random for practical purposes—depth estimation is infeasible for paraphrastic chains under these conditions.

Train → Test	Test set	F1
Continuation → Cont.	Continuation	0.72
Paraphrastic → Para.	Paraphrastic	0.60
Continuation → Para.	Paraphrastic	0.42
Para. → Continuation	Continuation	0.38

Table 7: Cross-transfer between chain types. Near-chance performance demonstrates no unified depth signal exists across generation regimes.

3.4 Experiment 4: Fine-Grained Ordinal Classification

We test whether depth discrimination extends beyond 3-class to full 5-class ordinal classification (gen-0 through gen-4).

Granularity	Method	Macro F1	MAE
3-class	XGBoost	0.72	0.31
5-class	XGBoost	0.38	0.95
5-class	CORAL	0.36	0.98
5-class	Frank-Hall	0.39	0.92

Table 8: Ordinal classification results. 5-class is infeasible regardless of method, with 34pp degradation from 3-class.

Depth	F1	Support	Confused with
gen-0	0.85	4950	gen-1 (10%)
gen-1	0.45	4900	gen-2 (28%)
gen-2	0.22	4600	gen-3 (35%), gen-1 (22%)
gen-3	0.18	4200	gen-2 (33%), gen-4 (30%)
gen-4	0.20	3800	gen-3 (38%), gen-2 (21%)

Table 9: Per-depth F1 for 5-class classification. Depths 2–4 are mutually indistinguishable (F1$\leq$0.22), with 72% of off-diagonal confusion concentrated in this block.

Five-class ordinal classification is infeasible: macro F1=0.38 with MAE=0.95. The performance degradation from 3-class (34pp drop) is catastrophic and consistent across methods—ordinal-specific approaches (CORAL, Frank-Hall) provide no benefit over standard XGBoost.

The per-depth breakdown reveals that classification performance is driven almost entirely by gen-0 detection (F1=0.85). Depths 2–4 are mutually indistinguishable: Cohen's $d = 0.08$ for entropy_mean and 0.12 for hapax_ratio between adjacent deep generations. This pattern is architecture-conditional: Pythia-held-out achieves 5-class F1=0.48, while OLMo and Qwen held-out drop to 0.28–0.31 (near 5-class random baseline of 0.20).

3.5 Experiment 5: End-to-End Neural Classifier

We evaluate whether a QLoRA fine-tuned Qwen3-0.6B model can learn depth representations beyond handcrafted features.

The QLoRA model achieves 5-class F1=0.44 (vs. XGBoost 0.38, +6pp), suggesting that learned representations capture some signal beyond handcrafted features. However, this improvement does not resolve the fundamental problem: LOGO generalization remains architecture-conditional (Pythia held-out 0.52, OLMo 0.35, Qwen 0.38).

The overfitting pattern is diagnostic: the model achieves train F1=0.62 but val F1=0.44 (gap=0.18), with the largest overfitting gaps on non-Pythia families (0.24–0.27). This indicates the model learns

Method	5-class F1	3-class F1	LOGO 5-class
XGBoost 7-feature	0.38	0.72	0.34
QLoRA Qwen3-0.6B	0.44	0.76	0.38
Δ	$+0.06$	$+0.04$	$+0.04$

Table 10: End-to-end vs. handcrafted features. QLoRA provides modest improvement but does not resolve the fundamental architecture-conditional limitation.

Held-out Family	5-class F1	Overfit gap
Pythia (2.8B + 1B)	0.52	0.10
OLMo-2-1B	0.35	0.27
Qwen (1.7B + 0.6B)	0.38	0.24

Table 11: QLoRA LOGO by held-out family. The overfitting gap (train F1 $-$ val F1) is large for non-Pythia families, indicating generator-specific memorization rather than generalizable depth learning.

generator-specific patterns rather than a universal depth representation—it memorizes per-generator artifacts that do not transfer across architectures. This directly supports our architecture-conditional framing.

4 Discussion

4.1 Why Depth Signal is Architecture-Conditional

Our results reveal two mechanisms driving the architecture-conditional pattern:

Reference model mismatch. Pythia-410M as the reference model creates a same-family advantage for Pythia generators. The reference model's vocabulary and training distribution align with Pythia generators, making entropy statistics sensitive to their depth-induced drift. For OLMo and Qwen—trained on different corpora with different tokenizers—the reference model's entropy measurements are noisy proxies at best.

Generator-specific drift dynamics. Even surface features (which do not depend on the reference model) show weaker depth correlations for non-Pythia generators (TTR ρ: Pythia -0.58, OLMo -0.35, Qwen3-0.6B -0.33). This suggests that some architectures produce more uniform lexical distributions across depths, compressing the dynamic range available for drift detection.

4.2 Signal Absence as Fingerprint

A generator's depth response profile—which features drift, which remain stationary, and by how much—encodes architectural information even when absolute depth estimation fails. The contrast between Pythia (strong drift, $\rho \approx -0.6$) and OLMo/Qwen (weak drift, $\rho \approx -0.2$) is itself discriminative: one could identify the generator family from the *pattern* of depth response without needing to solve the depth estimation problem.

This second-order signal has practical implications: in scenarios where generator identification is the goal (rather than depth estimation), measuring how a text's features respond to chain depth provides an orthogonal signal to traditional stylometric approaches.

4.3 Implications for Depth Estimation Research

Our findings establish several practical boundaries:

1. **Granularity ceiling**: $\leq$3 classes is the practical maximum for cross-generator depth estimation. Fine-grained ordinal approaches (5+ classes) require generator-specific models.
2. **Architecture scope**: depth estimation is feasible primarily within the same model family as the reference model. Cross-family generalization requires family-matched reference models or fundamentally different features.
3. **Process specificity**: depth models are not transferable across generation regimes (continuation vs. paraphrastic). Any practical system must identify the chain type before attempting depth estimation.

4.4 Limitations

Several limitations constrain the generalizability of our findings. First, we use only Pythia-410M as the reference model; family-matched reference models (e.g., OLMo-based for OLMo generators) might recover signal for non-Pythia families. Second, our generators are limited to $\leq$2.8B parameters—larger models may exhibit different depth dynamics. Third, we evaluate only base models; instruction-tuned models with their modified sampling distributions may show different patterns. Fourth, chain construction uses single-model continuation, while real-world contamination involves heterogeneous generator sequences.

5 Conclusion

We present the first systematic study of ordinal depth estimation in recursive LLM generation chains and demonstrate that depth signal is fundamentally **architecture-conditional**. Rather than a universal property of recursive generation, depth-correlated feature drift depends on the alignment between generator architecture, training data, and the reference model used for feature extraction. This finding has direct implications for data provenance research: depth estimation systems require architecture-aware design rather than one-size-fits-all approaches.

Our key quantitative findings: (1) Cross-generator LOGO binary F1 ranges from 0.57 (Qwen3-0.6B) to 0.80 (Pythia-1B) with per-generator std=0.11; (2) surface features provide statistically indistinguishable performance from the full 7-feature set in cross-family settings ($p = 0.12$); (3) 5-class ordinal estimation achieves only F1=0.38 with depths 2–4 mutually indistinguishable; (4) paraphrastic and continuation chains have reversed depth dynamics with cross-transfer F1=0.38–0.42.

We propose that the *pattern of signal presence and absence* constitutes a second-order generator fingerprint, opening a new direction for generator characterization research that leverages depth response profiles rather than absolute depth estimation.

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